

Lecture 5 - Screening I

Simple Models

ECON 403/503
Fall 2020



$$30 - P_H > 10 - P_L > 6 - P_L = 0 \geq 12 - P_H$$

$\uparrow$ $\uparrow$
 IC_C $Point \textcircled{1} : IR_N$ IC_N

$$IC_N: 6 - P_L \geq 12 - P_H$$

$P_H \uparrow$, all constraints are still met.

$$\begin{aligned} \rightarrow P_L &= 6 \\ 30 - P_H &= 10 - P_L \end{aligned} \left. \vphantom{\begin{aligned} \rightarrow P_L &= 6 \\ 30 - P_H &= 10 - P_L \end{aligned}} \right\} \begin{array}{l} \text{System of 2 eq.} \\ \text{in 2 variables} \end{array}$$

$$P_L = 6, P_H = 26$$

$$30 - P_H \geq 10 - P_L \quad (IC_C)$$

$$6 - P_L \geq 12 - P_H \quad (IC_N)$$

$$30 - P_H \geq 0 \quad (IR_C)$$

$$6 - P_L \geq 0 \quad (IR_N)$$

$$P_H - P_L \leq 20 \quad (IC_C) \quad \bullet$$

$$P_H - P_L \geq 6 \quad (IC_N) \quad \bullet$$

$$P_H \leq 30 \quad (IR_C) \quad \bullet$$

$$P_L \leq 6 \quad (IR_N) \quad \bullet$$

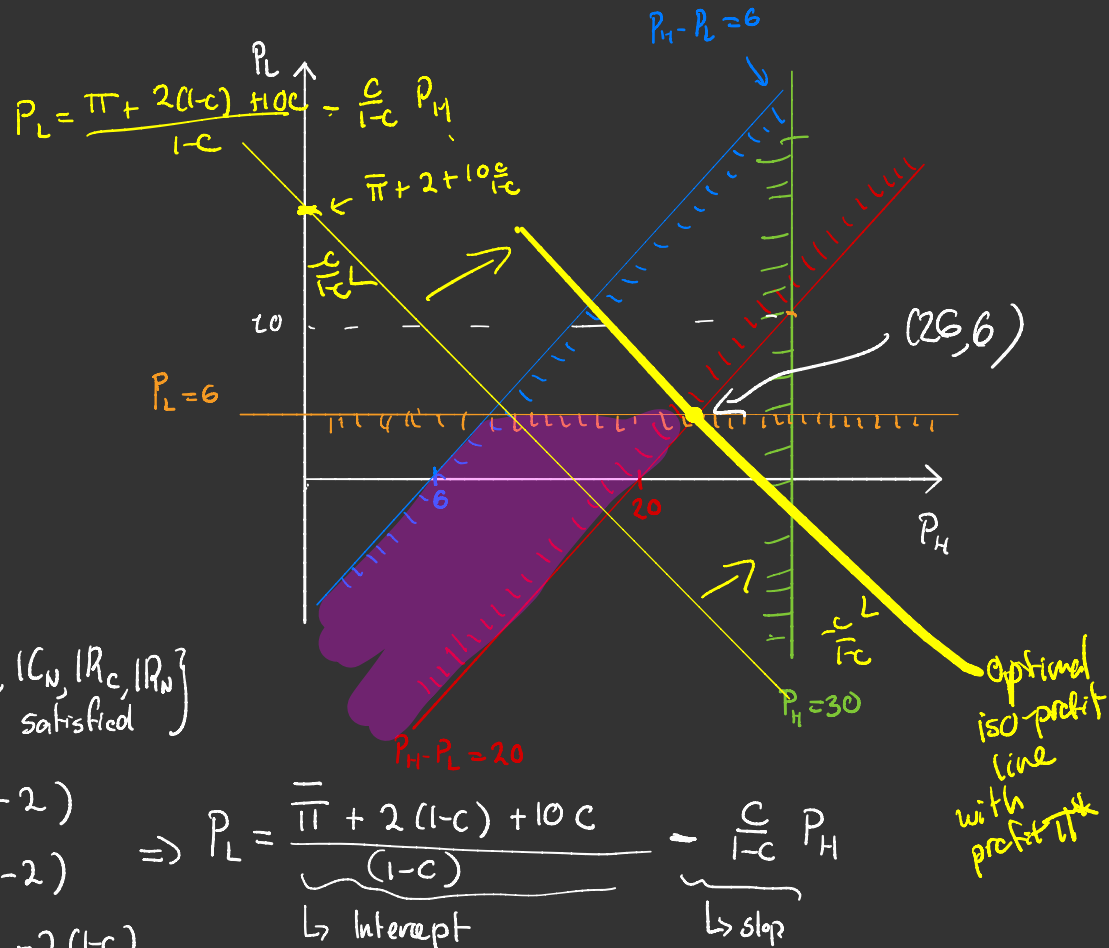
$$\text{Feasible set} = \{ (P_H, P_L) \in \mathbb{R}^2 \mid IC_C, IC_N, IR_C, IR_N \text{ are satisfied} \}$$

$$\text{Objective: } c(P_H - 10) + (1-c)(P_L - 2)$$

$$\text{Iso-profit line } \bar{\pi} = c(P_H - 10) + (1-c)(P_L - 2)$$

$$\bar{\pi} = cP_H - 10c + (1-c)P_L - 2(1-c)$$

$$\text{Optimal Profit } \pi^* = c(26 - 10) + (1-c)(6 - 2) = 16c + 4(1-c)$$



$$\Rightarrow P_L = \frac{\bar{\pi} + 2(1-c) + 10c}{(1-c)} - \frac{c}{1-c} P_H$$

$\hookrightarrow$ Intercept $\hookrightarrow$ slope

If $\bar{\pi} \uparrow$, intercept goes up

Profits under all schemes:

Complete Info: $20c + 4(1-c)$

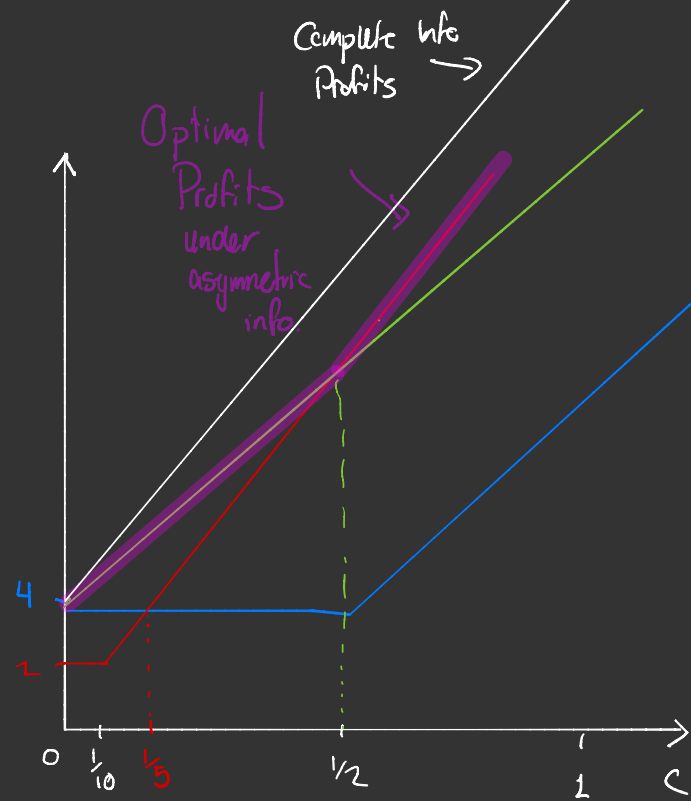
M Doesn't know types:

Low Quality only: 4 if $c \leq 1/2$
 $8c$ if $c > 1/2$

High Quality only: 2 if $c \leq 1/10$
 $20c$ if $c > 1/10$

Both Qualities: $16c + 4(1-c)$

Optimal Scheme is the upper envelope of the three profit functions $\Rightarrow$



If $c \leq 1/2$ offer to both
If $c > 1/2$ only High Quality

Rider Example

$$\begin{aligned} H: \quad \frac{\partial u_H}{\partial q} &= 3 \\ L: \quad \frac{\partial u_L}{\partial q} &= 2 \end{aligned} \left\{ \begin{array}{l} \text{Marginal value} \\ \text{for quality} \end{array} \right. \text{ is greater for type } H.$$

The problem becomes:

$$\max \quad \frac{1}{2} (2q_L - q_L^2) + \frac{1}{2} (3q_H - q_H^2)$$

$$q_H, q_L \geq 0$$

↳ When solving this problem
focus on the interior solution

↳ Can be guaranteed by using some assumptions on utilities for quality

First-Best of the monopolist:-

$$\max_x: E[\text{profits}]$$

$$(q_H, t_H, q_L, t_L)$$

subject IR_H, IR_L : Individual Rationality
Constraints only

At an interior solution, the FOCs must hold with equality

$$\left. \begin{aligned} \frac{\partial \text{Profits}}{\partial q_L} &= \frac{1}{2} (2 - 2q_L) = 0 \\ \frac{\partial \text{Profits}}{\partial q_H} &= \frac{1}{2} (3 - 2q_H) = 0 \end{aligned} \right\} \begin{aligned} q_L &= 1 \\ q_H &= \frac{3}{2} \end{aligned}$$

$$\text{Optimal Profits: } \frac{1}{2} (2 - 1) + \frac{1}{2} \left(\frac{9}{2} - \frac{9}{4} \right) = \frac{13}{8}$$

Second-Best (Asymmetric Information)

$$\max \quad \frac{1}{2} (t_L - q_L^2) + \frac{1}{2} (t_H - q_H^2)$$

$$(q_H, t_H), (q_L, t_L)$$

$$\text{subject to: } \left. \begin{array}{l} 3q_H - t_H \geq 3q_L - t_L \quad (IC_H) \\ 2q_L - t_L \geq 2q_H - t_H \quad (IC_L) \\ 3q_H - t_H \geq 0 \quad (IR_H) \\ 2q_L - t_L \geq 0 \quad (IR_L) \end{array} \right\} \mathcal{F} = \left\{ (q_H, t_H), (q_L, t_L) \mid \begin{array}{l} \text{both} \\ \text{menus satisfy } IC_H, IC_L, IR_H, IR_L \end{array} \right\}$$

Feasible set

What is the structure of the feasible set?

Indifference curves of both types of consumers
in (q, t) -space

Indifference curves in (q, t) space

$$H: \{(q, t) : u_H = 3q - t\}$$

$$L: \{(q, t) : u_L = 2q - t\}$$

↳ solve for t :

$$H: t = 3q - u_H$$

$$L: t = 2q - u_L$$

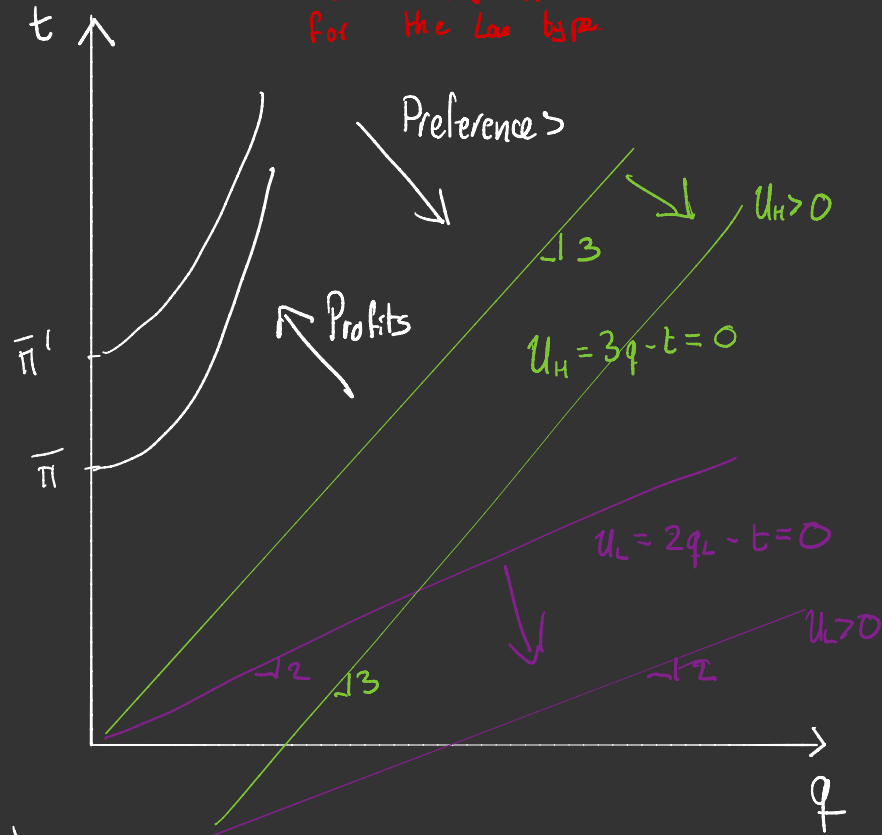
What happens when the utility of the consumers increase?

$$u_H, u_L \uparrow$$

$$\text{Expected profits are: } \frac{1}{2}(t_L - q_L^2) + \frac{1}{2}(t_H - q_H^2)$$

↑ with t , ↓ q

Single-Crossing Condition: At any point the indifference curve of the High type is steeper than the one for the Low type.



$$\text{Iso-profit line: } \{(q, t) \mid \bar{\pi} = t - q^2\} \Rightarrow t = \bar{\pi} + q^2$$

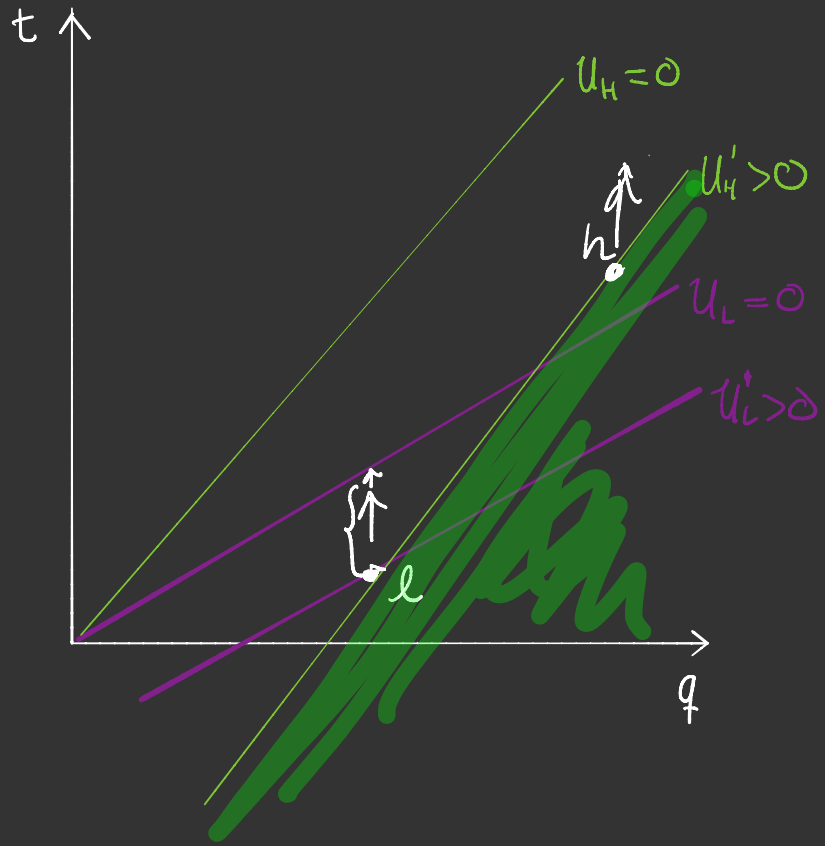
① IR_L holds with equality

$$|u_L = 2q_L - t_L = 0|$$

Towards a contradiction, suppose not, i.e., $u'_L > 0$, suppose that $(q_L, t_L) = l$, for IC_H to hold, H 's bundle, say

$h = (q_H, t_H)$ must be in the shaded region,

We can increase the transfers t_L, t_H , by the same small amount without violating the constraint $\rightarrow$



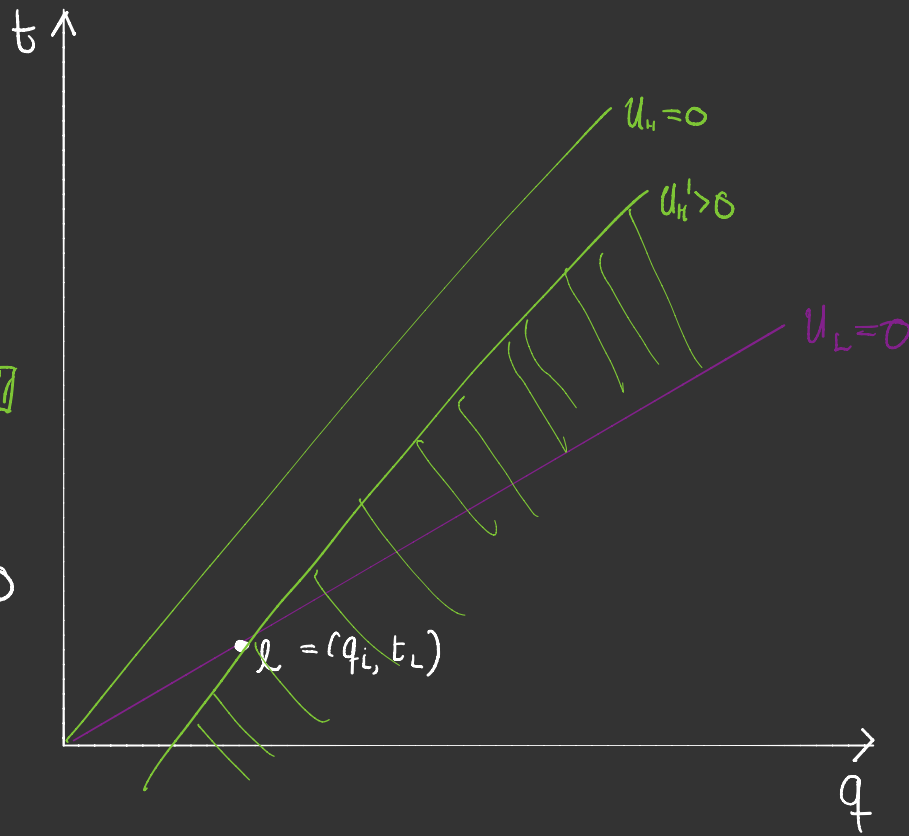
② We can ignore IR_H .

Since IR_L holds with equality
l needs to be in the
indifference curve with $u_L=0$,
to satisfy IC_H , h must
be in the shaded region

but any menu (q_H, t_H) in
the shaded region has $u_H' > 0$

$\Rightarrow IR_H$ holds:

$$3q_H - t_H > 0$$



⑤ IC_H held as an equality.

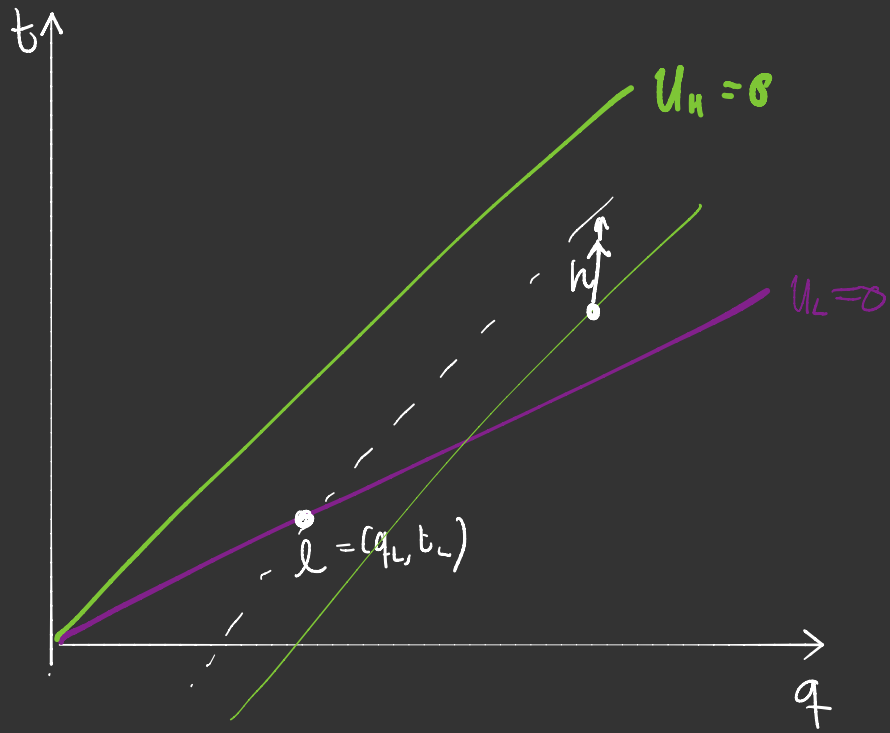
Suppose not, i.e.

$$\exists q_H - t_H > \exists q_L - t_L$$

$\Rightarrow (q_H, t_H)$ does not pass through
the point l , so say
is at point h .

But this means we can
increase t_H a little bit,
and increase the profit

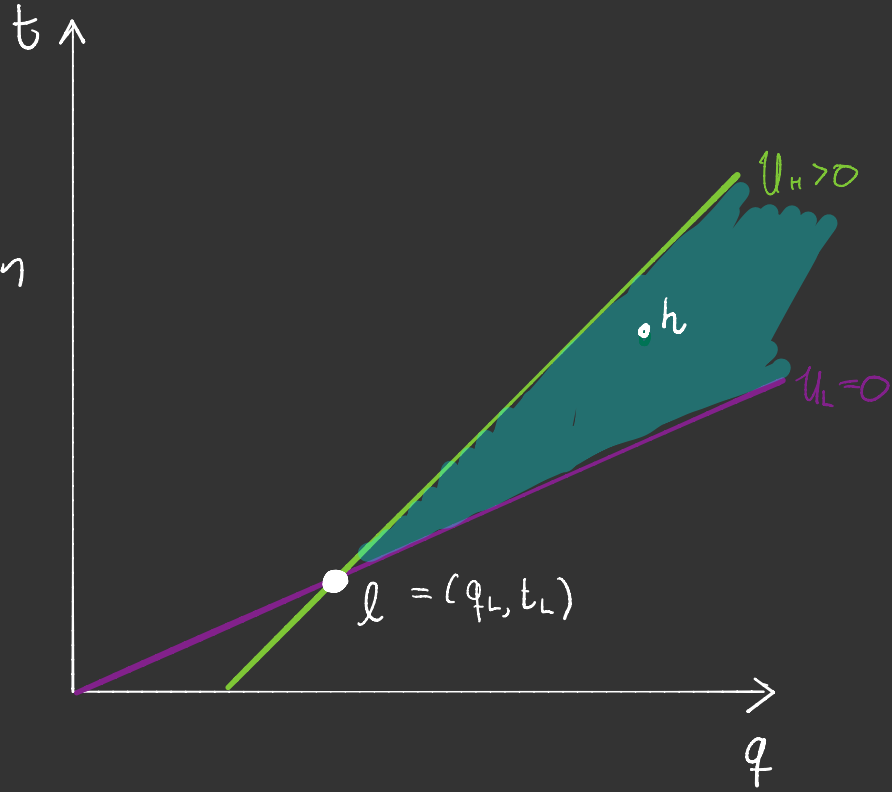
$$\therefore \exists q_H - t_H = \exists q_L - t_L$$



④ $q_H \geq q_L$

Suppose type L allocation is l . This type-H allocation must be in the shaded region i.e, h must be in

Therefore, h must be to the right of l and above u_L and u_H



⑤ We can ignore IC_L

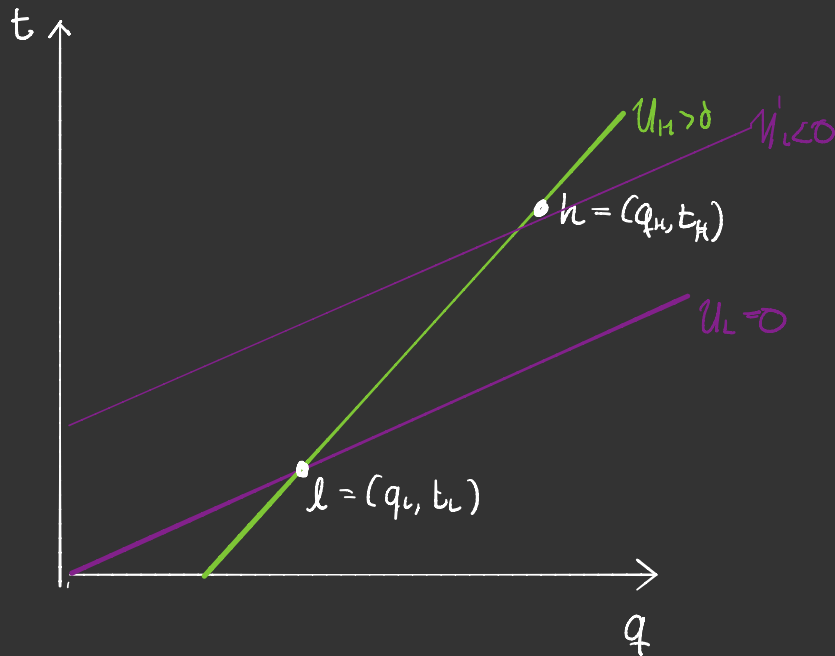
It follows from IC_H and IR_L

Holdeney with equality

↳ IC_H holds with equality

↳ $q_H \geq q_L$

$$2q_L - t_L \geq 2q_H - t_H \quad \checkmark$$



IC_H and IR_L hold with equality

$$\Rightarrow \left. \begin{array}{l} 3q_H - t_H = 3q_L - t_L \\ 2q_L - t_L = 0 \end{array} \right\} \begin{array}{l} \text{Solve for } t_L, t_H \text{ in terms} \\ \text{of } q_L, q_H \end{array}$$

$$\textcircled{1} \text{ If } 3q_H - t_H \geq 3q_L - t_L = 0 \Rightarrow 3q_H - t_H \geq 0 \quad IR_H$$

$\uparrow$
 IC_H

$$\textcircled{2} \quad 3q_H - t_H = 3q_L - t_L$$

$$\textcircled{3} \quad 3q_H - t_H \geq 3q_L - t_L \Rightarrow 3(q_H - q_L) \geq t_H - t_L$$

$$2q_L - t_L \geq 2q_H - t_H \Rightarrow t_H - t_L \geq 2(q_H - q_L)$$

$$\Rightarrow 2(q_H - q_L) \leq t_H - t_L \leq 3(q_H - q_L)$$

$$\Rightarrow 2(q_H - q_L) \leq 3(q_H - q_L)$$

$$\Rightarrow 0 \leq (3-1)(q_H - q_L) = q_H - q_L$$

$$\Rightarrow q_H \geq q_L$$

$$t_H - t_L \geq 2(q_H - q_L)$$

but this IC_L

$\textcircled{4}$ Using $\textcircled{2}$ and $\textcircled{3}$ we know that

$$t_H - t_L = 3(q_H - q_L) \geq 2(q_H - q_L)$$

$$\left. \begin{array}{l} t_L = 2q_L \\ 3q_H - t_H = 3q_L - t_L \end{array} \right\} \Rightarrow 3q_H - t_H = 3q_L - 2q_L$$

$$\Rightarrow 3q_H - 3q_L + 2q_L = t_H$$

Subs. into the objective

$$\underline{3q_H - q_L = t_H}$$

↙

$$\frac{1}{2} (3q_H - q_L - q_H^2) + \frac{1}{2} (2q_L - q_L^2)$$